

AMENDMENT NO. 6 NOVEMBER 2025
TO
IS 2553 (PART 2) : 2019 SAFETY GLASS — SPECIFICATION
PART 2 FOR ROAD TRANSPORT
(First Revision)

(Page 1, clause 3.3.5) — Insert the following at the end:

‘3.3.5.1 Treated Laminated Glass — When at least one of the layers of glass, of which it is composed, has been specially treated to increase its mechanical strength and to condition its fragmentation after shattering.’

[Page 3, clause 4.1.2.1.4 (see also Amendment No. 2)] — Insert the following at the end:

‘4.1.2.1.5 “III” for treated laminated glass.’

(Page 3, clause 4.1.2.2.2) — Substitute the following for the existing:

‘4.1.2.2.2 “XI” for laminated glass. In addition, the appropriate application will be signified by/D for laminated-glass panes with enhanced mechanical properties.’

Price Group 4

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[Page 4, Table 1 (see also Amendment No. 2)] — Substitute the following for the existing:

‘Table 1 Summary of Performance Requirements

(Clause 5)

SI No.	Test	Windscreens					Panes								
		Laminated Glass		Treated Laminated Glass		Glass Plastics	Uniformly Toughened Glass		Laminated Glass			Double Glazed Unit ¹⁾	Glass Plastics	Rigid Plastic	Flexible Plastic
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
i)	Marking	II	II/P	III	III/P	IV		P	XI	XI/D	XI/P	VI	XII	VIII	IX
ii)	Light transmittance	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1
iii)	Resistance to abrasion	5.1.2	5.1.2	5.1.2	5.1.2	5.1.2	—	5.1.2	5.1.2	5.1.2	5.1.2	—	5.1.2	5.1.2	—
iv)	Resistance to temperature changes	—	5.2.1	—	5.2.1	5.2.1	—	5.2.1	—	—	5.2.1	—	5.2.1	—	—
v)	Resistance to fire	—	5.2.2	—	5.2.2	5.2.2	—	5.2.2	—	—	5.2.2	—	5.2.2	—	—
vi)	Resistance to chemicals	—	5.2.3	—	5.2.3	5.2.3	—	5.2.3	—	—	5.2.3	—	5.2.3	—	—
vii)	Resistance to radiation	5.3.1	5.3.1	5.3.1	5.3.1	5.3.1	—	5.3.1	5.3.1	5.3.1	5.3.1	—	5.3.1	—	—
viii)	Resistance to high temperature	5.3.2	5.3.2	5.3.2	5.3.2	5.3.2	—	5.3.2	5.3.2	5.3.2	5.3.2	—	5.3.2	—	—
ix)	Resistance to humidity	5.3.3	5.3.3	5.3.3	5.3.3	5.3.3	—	5.3.3	5.3.3	5.3.3	5.3.3	—	5.3.3	5.3.3	
x)	Optical distortion	5.4.1	5.4.1	5.4.1	5.4.1	5.4.1	—	—	—	—	—	—	—	—	—

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Sl No.	Test	Windscreens					Panels								
		Laminated Glass		Treated Laminated Glass		Glass Plastics	Uniformly Toughened Glass		Laminated Glass			Double Glazed Unit ¹⁾	Glass Plastics	Rigid Plastic	Flexible Plastic
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
xi)	Image separation	5.4.2	5.4.2	5.4.2	5.4.2	5.4.2	—	—	—	—	—	—	—	—	—
xii)	Fragmentation	—	—	6.2.4	6.2.4	—	5.5.1.1	5.5.1.1	—	—	—	—	—	—	—
xiii)	Head-form ^{2) 3)}	5.4.3	5.4.3	5.4.3	5.4.3	5.4.3	—	—	—	5.5.2.2 ³⁾	—	5.5.3.2	—	5.6.1	—
xiv)	2260 g ball	5.4.4	5.4.4	5.4.4	5.4.4	5.4.4	—	—	—	5.5.2.3 ³⁾	—	—	—	—	—
xv)	227 g ball	5.4.5	5.4.5	5.4.5	5.4.5	5.4.5	5.5.1.2	5.5.1.2	5.5.2.1	5.5.2.1	5.5.2.1	5.5.2.1	5.5.2.1	5.6.2	5.7.1 and 5.7.2
xvi)	Flexibility and fold	—	—	—	—	—	—	—	—	—	—	—	—	5.6.3	5.7.3
xvii)	Weathering	—	—	—	—	—	—	—	—	—	—	—	—	5.6.4	5.7.4
xviii)	Cross-cut	—	—	—	—	—	—	—	—	—	—	—	—	5.6.5	—

¹⁾ Each component pane shall satisfy the appropriate tests for the type of glazing.

²⁾ This test is optional for 'L' category of vehicles.

³⁾ This test shall only be carried out on laminated-glass panes bearing the additional symbol/D.'

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(Page 7, clause 5.4.5.5) — Insert the following at the end:

‘5.4.6 Requirements Applicable to Treated Laminated-Glass Windscreens

In the case of treated laminated-glass windscreens, additional fragmentation test to be conducted according to test procedure 6.2.4, tests such as head form and test of optical qualities may be carried out on a complete windscreen and remaining tests may be conducted on samples and/or flat test pieces which are specially made for the purpose. One test piece shall be tested for the specified point of impact. The test piece shall be as described in 6.2.4.2.’

(Page 7, clause 5.5.2.1.4) — Insert the following at the end:

‘5.5.2.2 Head form Test

The provisions concerning the head form test shall apply to laminated-glass panes bearing the additional symbol/D:

- a) *Number of test pieces* — Eight flat test pieces measuring (1 100 mm × 500 mm) +10/-2 mm shall be subjected to testing;
- b) *Method of test:*
 - 1) The method used shall be that described in 6.5; and
 - 2) The height of drop shall be 1.50 m +0/-5 mm.
- c) *Interpretation of results:*
 - 1) This test shall be deemed to have given a satisfactory result if the following conditions are met:
 - i) The test piece yields and breaks, displaying numerous circular cracks centered approximately on the point of impact;
 - ii) Tears in the interlayer are allowed, but the manikin's head shall not pass through; and
 - iii) No large fragments of glass shall become detached from the interlayer.
 - 2) A set of test pieces submitted for approval shall be considered satisfactory from the point of view of the head form test if at least seven of the eight test pieces meet the requirements.

‘5.5.2.3 2.260 g Ball test

The provisions concerning the 2.260 g ball test shall apply to laminated-glass panes bearing the additional symbol/D;

- a) Twelve square test pieces of 300 mm side shall be subjected to testing.
- b) *Method of test:*
 - 1) The method used shall be that described in 6.4; and
 - 2) The height of drop (from the under-face of the ball to the upper-face of the test piece) shall be 4 m +25/-0 mm.
- c) *Interpretation of results:*
 - 1) The test shall be deemed to have given a satisfactory result if the ball does not pass through the glazing within five seconds after the moment of impact; and
 - 2) A set of test pieces submitted for approval shall be considered satisfactory from the point of view of the 2.260 g ball test if at least eleven of the twelve tests have given a satisfactory result.’

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(Page 8, clause **5.6.2.2**) — Substitute the following for the existing:

‘**5.6.2.2** Ten flat square pieces (300 +10/-0 mm) or ten substantially flat finished parts or specially prepared or cut from the flattest part of a safety glass.’

(Page 9, clause **5.6.3.1**) — Substitute the following for the existing:

‘**5.6.3.1** One flat test piece substantially flat or specially prepared or cut from the flattest part of a safety glass measuring 300 mm × 25 mm shall be subjected to testing.’

(Page 9, clause **5.6.4.3**) — Substitute the following for the existing:

‘**5.6.4.3** Three flat pieces 130 mm × 40 mm cut from a flat sheet sample, or substantially flat or specially prepared Sample shall be subjected to testing.’

(Page 10, clause **6.2.3.1**) — Insert the following at the end:

6.2.4 Fragmentation Test for Treated Laminated Glass-Windscreen

6.2.4.1 Method of test

- a) Before each test, the test pieces or samples shall be stored for not less than four hours at a temperature of $23\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$. The tests shall take place as soon as possible after the test pieces or samples have been taken out of the receptacle in which they were stored;
- b) The glass pane to be tested shall not be rigidly secured; it may however be fastened on an identical glass pane by means of adhesive tape applied all-round the edge;
- c) To obtain fragmentation, a hammer of about 75 g (*see fig. below*) or some other appliance giving equivalent results shall be used. The radius of curvature of the point approximately $0.2\text{ mm} \pm 0.1\text{ mm}$;



- d) One test shall be carried out at each prescribed point of impact;
- e) The examination of the fragments shall be made using any method validated on its accuracy of the counting itself and on its ability to find the correct location where the minimum and the maximum counting shall be done; and
- f) Permanent recording of the fragmentation pattern shall start within 10 s and shall end within 3 min after the impact. The technical service shall keep the permanent recordings of the fragmentation pattern

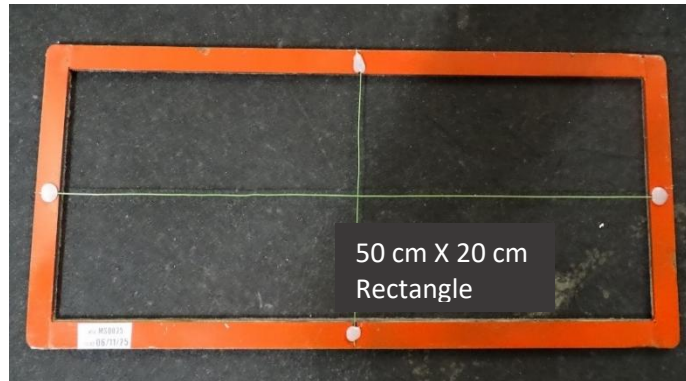
6.2.4.2 Test pieces

For treated laminated-glass windscreens one test piece measuring $(1\ 100\text{ mm} \times 500\text{ mm}) \frac{+5}{-2}\text{ mm}$ specially made or cut from the flattest part of a windscreen or one actual windscreen for the point of impact shall be subjected to testing.

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6.2.4.3 Interpretation of Results

- a) For each point of impact, the fragmentation test shall be considered to have given a satisfactory result if the total surface area of fragments having a surface area of more than 2 cm² comprised in a rectangle at least 20 cm high and 50 cm long (*see* fig. below), represents not less than 15 percent of the surface of that rectangle:



- 1) In the case of a sample:
 - i) For vehicles of category M1, the centre of the rectangle shall be situated within a circle having a radius of 10 cm centred on the projection of the middle of segment V1 V2;
 - ii) For vehicles of categories M and N other than M1, the centre of the rectangle shall be situated within a circle having a radius of 10 cm centred on the projection of point 0;
 - iii) For agricultural and forestry tractors and for construction-site vehicles, the position of the visibility zone shall be indicated in the test report; and
 - iv) The height of the above rectangle may be reduced to 15 cm for windscreens which are less than 44 cm high or whose angle of installation is less than 15° from the vertical; the percentage of visibility shall at least be equal to 10 percent of the area of the corresponding rectangle.
 - 2) In the case of a test piece, the centre of the rectangle shall be situated on the greater axis of the test piece at 450 mm from one of its edges.
- b) The test piece(s) or sample(s) submitted for approval shall be considered satisfactory from the point of view of fragmentation if either of the following conditions is met:
- 1) The test gives a satisfactory result for each point of impact; or
 - 2) The test having been repeated on a new set of four test pieces for each point of impact for which it had originally given an unsatisfactory result, the four new tests, performed at the same impact points all give a satisfactory result.'

(Page 16, clause 6.3.3.1) — Substitute the following for the existing:

‘6.6.3.1 The test pieces shall be either flat samples or specially prepared or cut from the flattest part of a safety glass measuring 100 mm × 100 mm having both surfaces substantially plane and parallel and having a fixing hole 6.4 mm + 0.2 mm in diameter drilled in the centre, if necessary.’

(Page 17, clause 6.11.3.1) — Substitute the following for the existing:

‘6.11.3.1 Test pieces shall be either flat samples or finished products or substantially flat or prepared sample.’

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(Page 29, clause **6.15.3.1**) — Substitute the following for the existing:

‘6.15.3.1 The test pieces shall be either flat samples or substantially flat or specially prepared or cut from the flattest part of a safety glass measuring 180 mm × 25 mm.’

(Page 33, clause **6.17.2**) — Substitute the following for the existing:

‘6.17.2 Method of Test

From the material of the nominal thickness a rectangular flat sample 300 mm long and 25 mm wide specially prepared or cut from the flattest part of a safety glass and horizontally clamped into a clamping device in such a way that 275 mm of the length of the sample freely extend over the holding device. This free end shall be supported horizontally by an appropriate device until the test begins. Sixty seconds after removal of this support, the vertical deviation of the free end is indicated in mm. If this deviation exceeds 50 mm a 180° fold test is performed subsequently. The sample is folded concisely; after that it is folded round 0.5 mm thick piece of sheet metal in such a way that it tightly contacts it on both sides.’

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[Page 43, clause **11.1** (see also Amendment No. 4)] — Substitute the following for the existing:

‘Table 6 Conformity of Production Test

(Clause 11.1)

Sl No.	Test	Windscreens					Panels						
		Laminated Glass		Treated Laminated Glass		Glass Plastics	Uniformly Toughened Glass		Laminated Glass			Double Glazed unit ¹⁾	Glass Plastics
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	Marking	II	II/P	III	III/P	IV		P	XI	XI/D	XI/P	VI	XII
ii)	Light transmittance	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1	5.1.1
iii)	Resistance to high temperature/boil test	5.3.2	5.3.2	5.3.2	5.3.2	5.3.2	—	5.3.2	5.3.2	5.3.2	5.3.2	—	5.3.2
iv)	Optical distortion	5.4.1	5.4.1	5.4.1	5.4.1	5.4.1	—	—	—	—	—	—	—
v)	Secondary image separation	5.4.2	5.4.2	5.4.2	5.4.2	5.4.2	—	—	—	—	—	—	—
vi)	Fragmentation	—	—	6.2.4	6.2.4	—	5.5.1.1	5.5.1.1	—	—	—	—	—
vii)	Head-form test	5.4.3	5.4.3	5.4.3	5.4.3	5.4.3	—	—	—	—	—	5.5.3.2	—
viii)	2 260 g ball	5.4.4	5.4.4	5.4.4	5.4.4	5.4.4	—	—	—	5.5.2.3	—	—	—

¹⁾ Each component pane shall satisfy the appropriate tests for the type of glazing.’